# Introduction To Web Technologies

Chapter 1

# Outlines

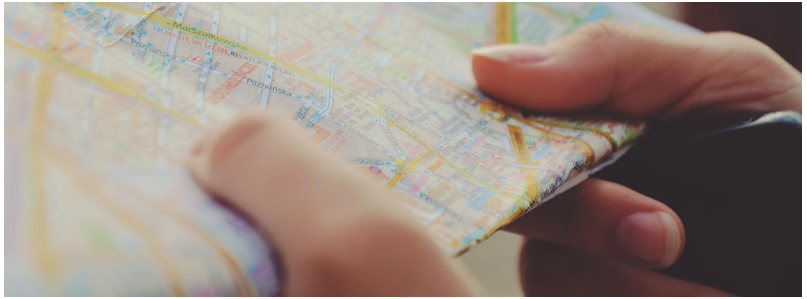
## Introduction
Internet and its application, service provider, DNS

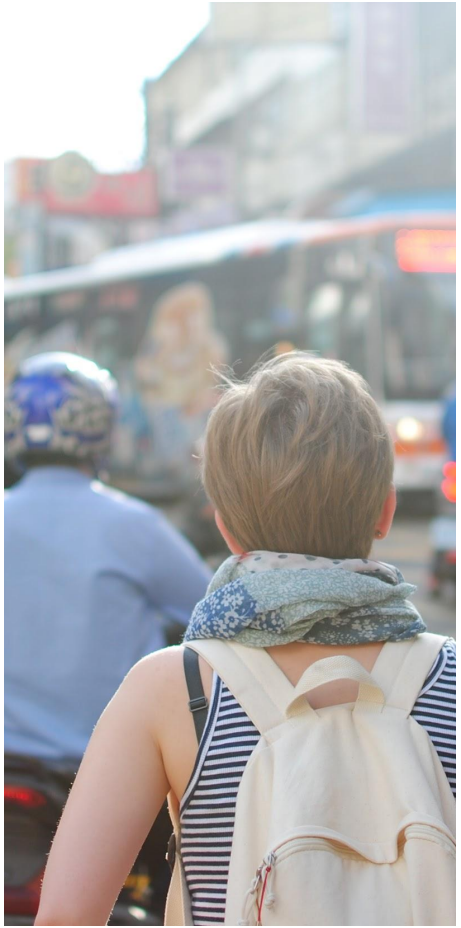
## Protocols
HTTPS, HTTP, SMTP, FTP, UDP, TCP/IP

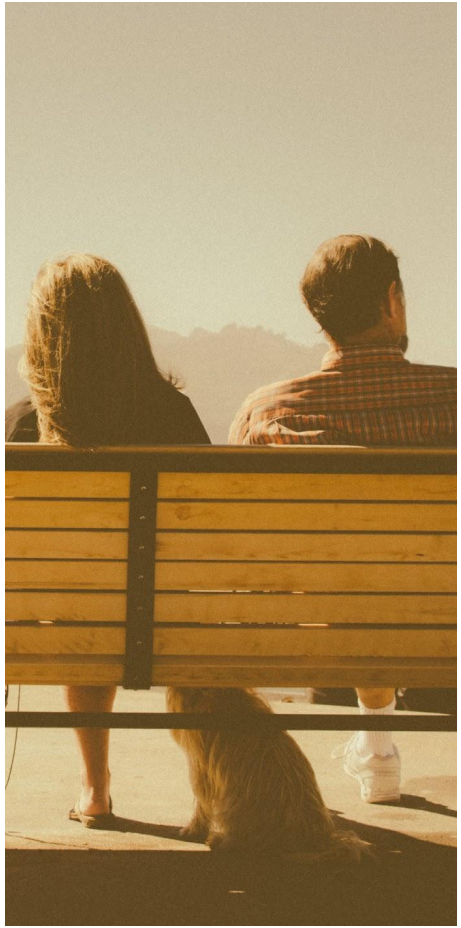
## DNS
DNS hierarchy, DNS registration issues

## Server
Client server model, cross browser communication

## URL
Absolute and relative url







# Evolution of Web

## Web1.0, Web 2.0, web3.0, web4.0

Web 1.0: The Static Web and used HTML, HTTP and Browsers. (1990s – Early 2000s)

Web 2.0: The social and interactive web. Used HTML, CSS, Javascript and AJAX, API, Cloud, responsive web (Early 2000s – Present)

Web 3.0: The Decentralized & Intelligent Web. used Blockchain & Cryptography Smart Contracts AI & Machine Learning and Edge & Cloud Computing

Web 4.0: Future of the Web, Intelligent Web and emotional web, Fully AI driven.

# History of web

- 1960s: The Idea of a Computer Network
- 1970s: The Birth of the Internet
- 1989: Tim Berners-Lee Proposes the Web
- 1990: The First Website
- 1991: The Web Goes Public
- 1993: The First Web Browser – Mosaic
- 1994 – 1999: The Dot-Com Boom
- 2000: Dot-Com Bubble Bursts
- 2004 – 2005: Rise of Social Media
- 2010s: AI, Blockchain & Privacy Concerns
- 2020s: The Future of the Web
- AI-Powered Assistants: ChatGPT, Google Bard.
- Decentralized Internet: blockchain-based applications.
- Metaverse & AR/VR: Facebook rebrands as Meta to focus on virtual worlds.
- Quantum Computing: Companies explore next-gen web infrastructure.

# WWW

WWW stands for World Wide Web and is commonly known as the Web. The WWW was started by CERN in 1989.  WWW is defined as the collection of different websites around the world, containing different information shared via local servers(or computers).

# Search Engine

- A search engine is a software system that helps people find information
- Features of Search Engine:
  - Crawling: Visit the web pages
  - Indexing: save those visited data
  - Ranking: using algorithm rank them
- Examples are
  - Google
  - Bing
  - Yahoo
  - Baidu
  - Yandex

# Website and web application:

| **Website** | **Web Application** |
|---|---|
| A collection of interlinked static or dynamic web pages accessible through a browser. | A software application that runs on a web server and is accessed through a browser. |
| To present information (read-only or mostly one-way communication). | To allow user interaction, input, and processing (two-way communication). |
| News sites, blogs, portfolios, company websites. | Gmail, Facebook, Online Banking, Google Docs, WooCommerce store. |
| Focuses on Content | Focuses on functionality |
| Easier to build | Requires programming logic, time taken |
| | |

# URL = Uniform Resource Locator

- A URL is a Unique identifier that is contained by all the resources available on the internet.
- It can help to locate a particular resource due to its uniqueness.
- It is also known as the web address.
- A URL consists of different parts like protocol, domain name, etc.
- The users can access the URLs by simply typing them inside the address bar or by clicking any button or link web page.

# Parts of URL

https://www.example.com/webtech/javascirpt-basics/?id=2#what-is-js

- **The protocol or scheme:**
  A URL starts with a protocol that is used to access the resource on the internet. The resource is accessed through the Domain Name System or DNS. There are multiple protocols avaiable to use like HTTP, HTTPS, FTP, mailto, TELNET etc. The protocol used in the above URL is https.
- **Domain or Host Name:**
  It is the reference or name of the page that you are going to access on the internet. In this case, the domain name is: www.example.com.
- **Port Name:**
  It is defined just after the domain name by using the colons between itself and the domain name. Generally, it is not visible in the URL. The domain name and the port name combinely can be known as Authority. The default port for web pages is port80 (:80).

# Parts of URL

https://www.example.com/webtech/javascirpt-basics/?id=2#what-is-js

- **Path:**
  It refers to the path or location of a particular file or page stored on the web server to access the content of it. The path used here is: javascript-basics.
- **Query:**
  A query mainly found in the dynamic pages. It consists of a question mark(?) followed by the parameters. In above URL query is: ?.
- **Parameters:**
  These are the pieces of information inside a query string of URL. Multiple parameters can be passed to a URL by using the ampersand(&) symbol to separate them. The query parameter in above URL is: id=2.
- **Fragments:**
  The fragments appear at the end of a URL starts with a Hashtag(#) symbol. These are the internal page references that refers to a specific section within the page. The fragment in the above URL is: #what-is-js.
-

# Client Server Architecture

- Client-server architecture is a fundamental concept in system design where a network involves multiple clients and a server.
- Clients are devices or programs that request services or resources, while the server is a powerful machine providing these resources or services.
- This architecture allows efficient data management and resource sharing, making it popular in web applications, databases, and other network-based systems.
- By separating roles and distributing tasks, client-server architecture enhances performance, scalability, and security.
- Communication between clients and the server follows a request-response protocol, such as HTTP/HTTPS for web services or SQL for database queries.
- This design is prevalent in various applications, including web services, database management, and email systems, providing a robust framework for developing and managing complex, distributed systems efficiently.

# Importance in Client server Design

- **Centralized Management:** This architecture simplifies maintenance, updates, and security management.
- **Scalability:** As the number of clients grows, additional servers can be added, or existing server capacities can be expanded to handle increased demand.
- **Resource Optimization:** Servers are designed to handle intensive processing and large data storage, while clients are optimized for user interactions and requests.
- **Reliability and Availability:** Redundancies, backups, and load balancing techniques can be implemented on the server side to minimize downtime and ensure continuous service availability.
- **Enhanced Security:** Sensitive data can be securely stored on servers, and access can be tightly controlled and monitored. Encryption and authentication mechanisms can be more effectively implemented.

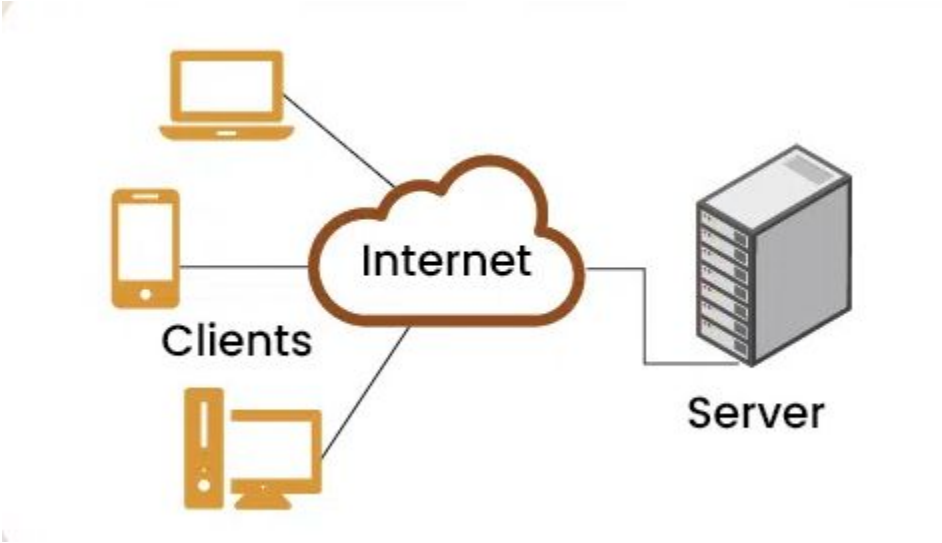
# Component of Client server Architecture

- **Client:** The client is a device that requests services or resources from the server. It initiates communication with the server and processes the results returned by the server.
- **Server:** The server is a device that provides services to clients. It processes client requests, performs the necessary computations, and sends back the appropriate responses.
- **Network:** The network facilitates communication between clients and servers. It enables data exchange using various protocols and ensures reliable connectivity.
- **Protocols:** Protocols are standardized rules that govern data transmission and communication between clients and servers. They ensure that data is transmitted in a secure, reliable, and understood manner.

# Component of Client server Architecture

- **Middleware:** Middleware is software that acts as an intermediary between clients and servers. It provides additional functionalities like authentication, logging, message queuing, and transaction management.
- **Database:** A database is a structured collection of data stored on the server. It stores and manages data that clients request, ensuring data consistency, integrity, and security.
- **User Interface (UI):** The UI is the part of the client application that interacts with the user. It provides a means for users to input data and view results returned by the server.
- **Application Logic:** Application logic refers to the code and algorithms that define the application's functionality. It processes user inputs, interacts with the server, and manages the flow of data between the client and the server.

# Client Server Architecture



# Types of client server architecture

- **1-tier architecture:** The presentation, business logic, and data layers are all stored on a single device or a shared storage device.
- **2-tier architecture:** The presentation and business logic layers are stored on the client while the data layer is stored on a server.
- **3-tier architecture:** The presentation layer is stored on the client, the business logic layer is stored on one server, and the data layer is stored on another server.
- **N-tier architecture:** The presentation and data layers are left untouched, compared to the 3-tier architecture. The difference is that N-tier architecture splits the business logic layer into multiple layers to improve performance, management, and stability.

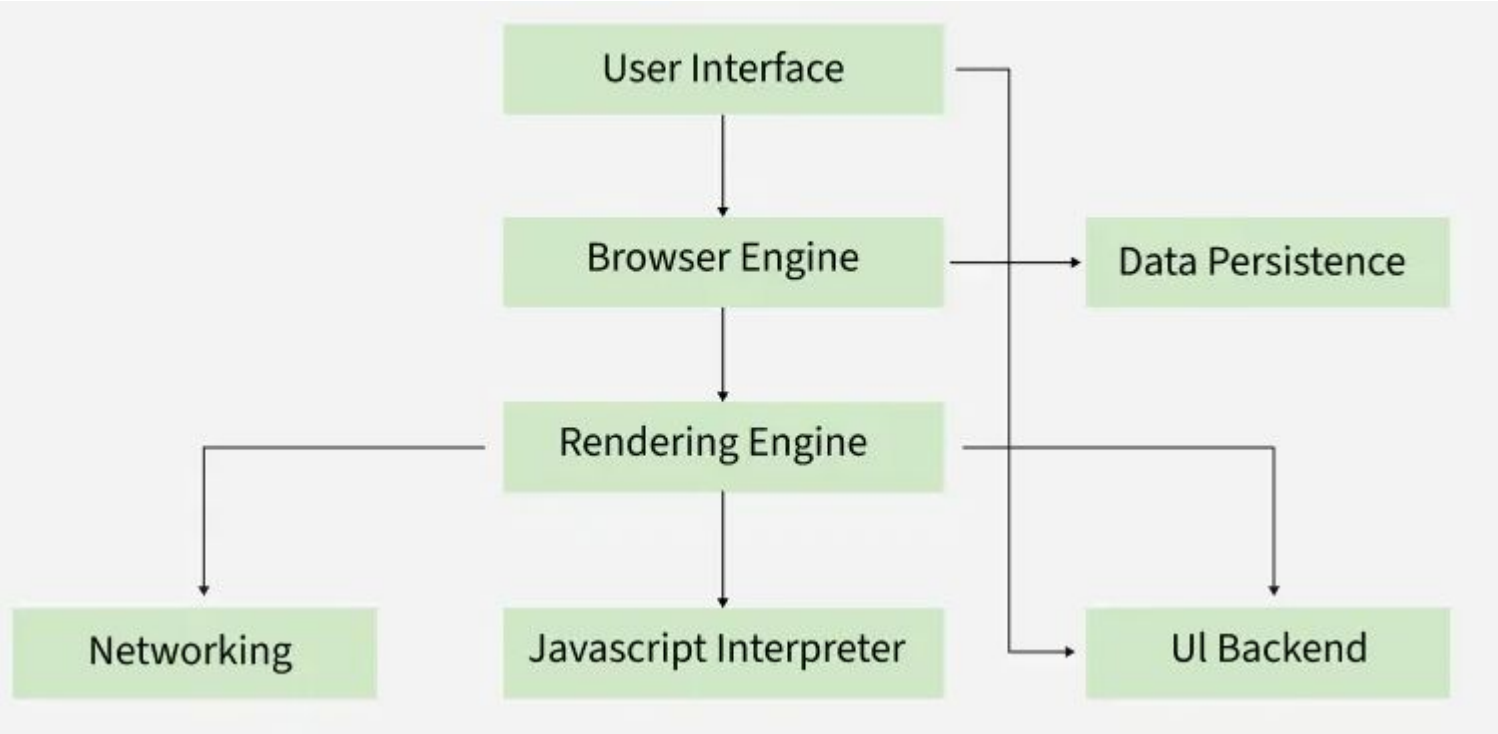
# Web Browser

- The web browser is an application software used to explore the World Wide Web (WWW).
- It acts as a platform that allows users to access information from the Internet by serving as an interface between the client (user) and the server.
- The browser sends requests to servers for web documents and services, then renders the received HTML content, including text, images, links, styles, and scripts.

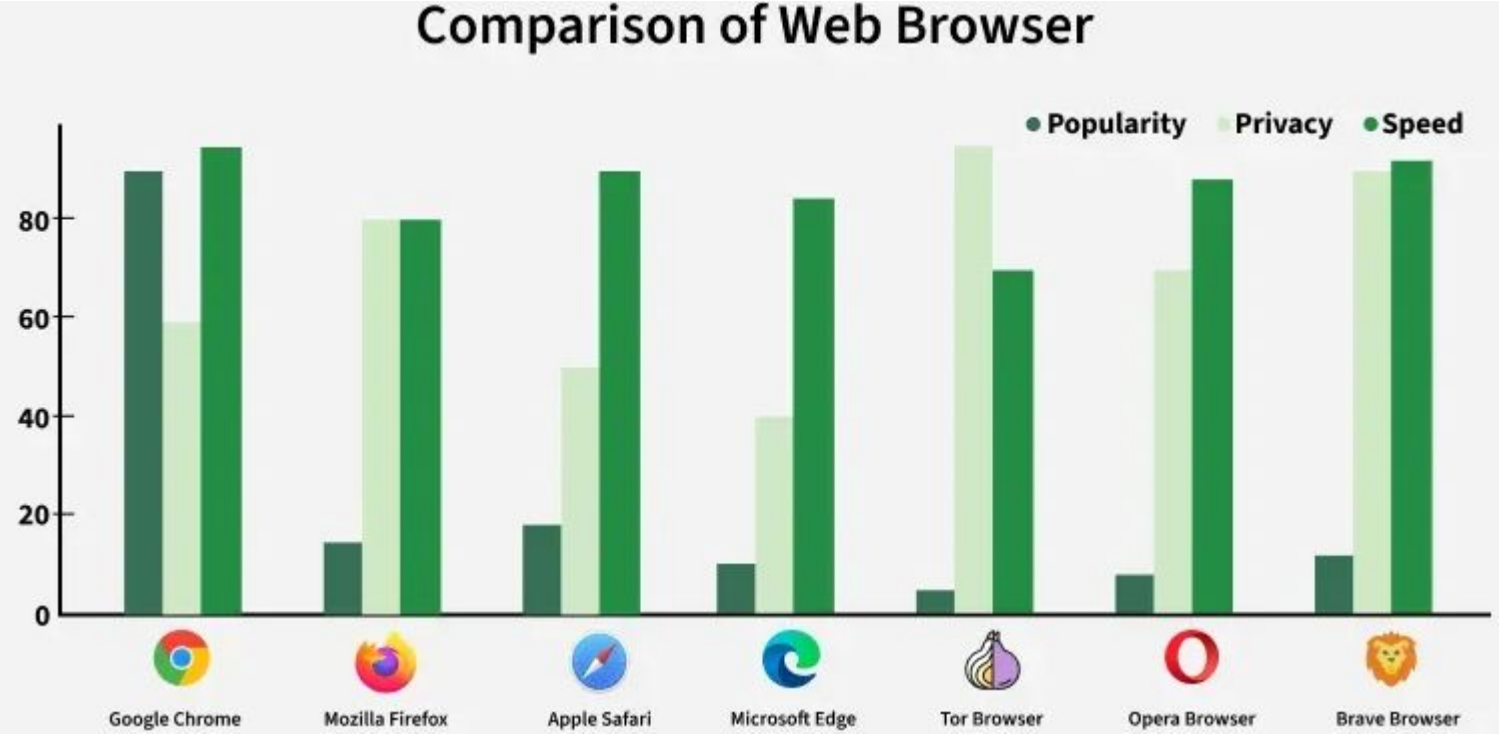
# History of the Web Browsers

- The first web browser, World Wide Web, was invented by Tim Berners-Lee in 1990 and later renamed Nexus.
- In 1993, Mark Andreessen and his team developed Mosaic, the first browser to display text and images together.
- He later created Netscape in 1994. Microsoft introduced Internet Explorer in 1995, bundled with Windows.
- This led to the development of other browsers like Mozilla Firefox, Google Chrome, Safari, and Opera, each offering unique features.

# Core Architecture of Web Browser



# Some Popular Web Browsers



# Cross Browser compatibility

Cross-browser compatibility refers to the ability of a

- website,
- web application,
- or feature to function

correctly and consistently across different web browsers and their various versions.

**(like Chrome, Firefox, Safari, Edge, Opera, etc.)**

# Why we need Cross Browser compatibility

- Different browsers interpret HTML, CSS, and JavaScript slightly differently.
- Without testing or proper coding practices, our site(web app) might look broken or behave unexpectedly for users on other browsers or devices.
- Cross-browser issues often arise from vendor-specific implementations, unsupported features, or old browser versions.
- We need to test below things
    - HTML/CSS rendering
    - Javascript rendering
    - Responsive design
    - Performance

# Web Server

- A web server is a software or hardware system that stores, processes, and delivers web content to users over the internet using the HTTP or HTTPS protocol.
  - The browser sends a request to the web server.
  - The server finds the requested file .
  - The server processes it and sends the result back to your browser.
  - Your browser displays the page for you to see.

# Examples of Web Server software

- Apache HTTP Server
- Nginx
- Microsoft IIS (Internet Information Services)

It works as follow

1. We enter www.nast.edu.np in our browser.
2. The browser sends an HTTP request to the server hosting that site.
3. The server receives the request, processes it, and locates the required file.
4. The server sends back an HTTP response containing the requested web page.
5. Our browser renders and displays the page to us.

# Network Protocols

Network protocols are a set of rules that are responsible for the communication of data between various devices in the network.

These protocols define guidelines and conventions for transmitting and receiving data, ensuring efficient and reliable data communication.

# Types of Network Protocols

- Network Communication
- Network Management
- Network Security

# Network Communication

Communication protocols are really important for the functioning of a network.

They are so crucial that it is not possible to have computer networks without them. These protocols formally set out the rules and formats through which data is transferred.

# Network Communication

- Hypertext Transfer Protocol(HTTP)
- Transmission Control Protocol(TCP)
- User Datagram Protocol(UDP)
- Border Gateway Protocol(BGP)
- Address Resolution Protocol(ARP)
- Internet Protocol(IP)
- Dynamic Host Configuration Protocol(DHCP)

# Network management

Network management protocols can also be used for troubleshooting connections between a host and a client.

- Gopher
- File Transfer Protocol(FTP)
- Post Office Protocol(POP3)

# Network Security

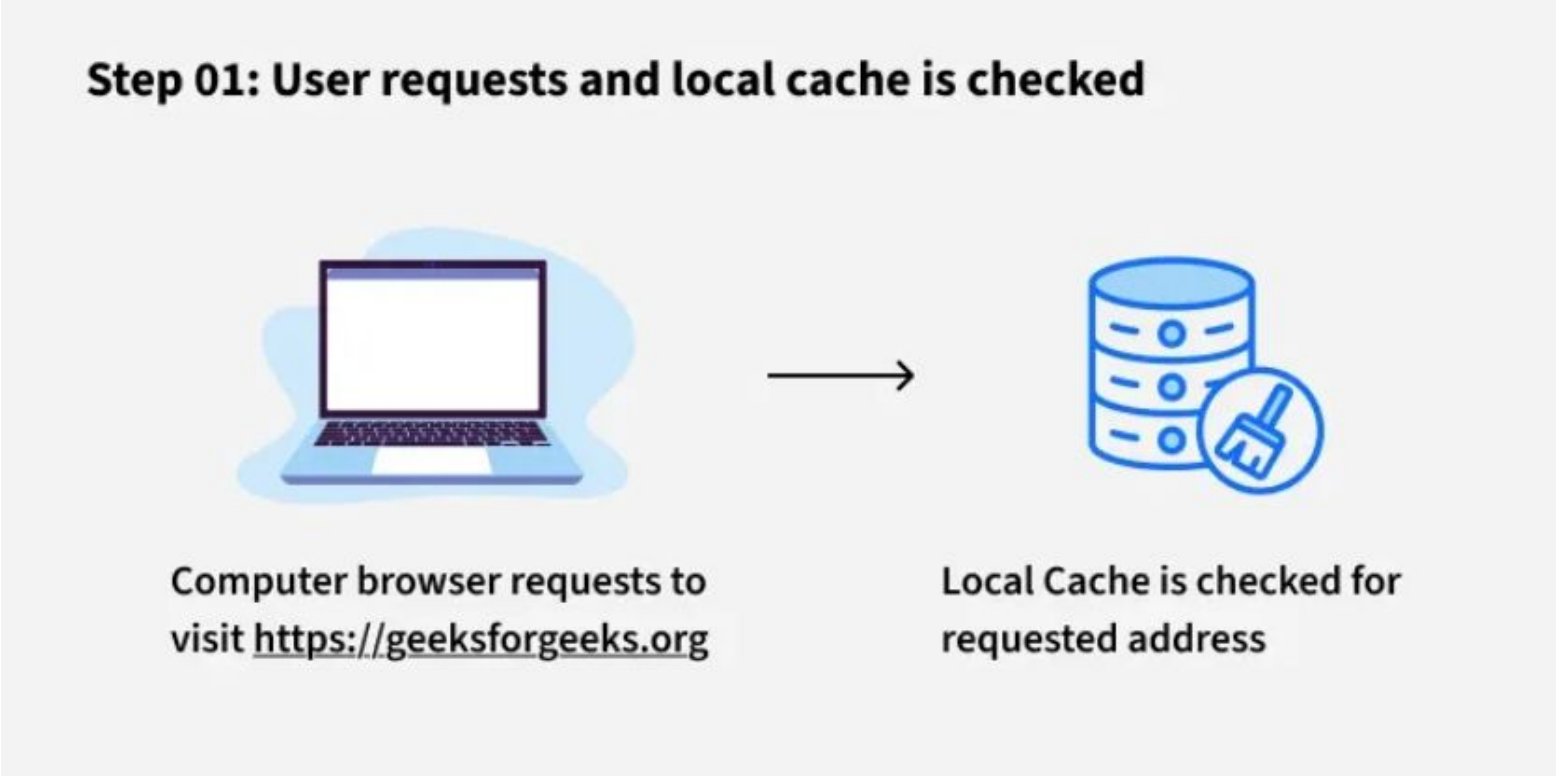
These protocols secure the data in passage over a network. These protocols also determine how the network secures data from any unauthorized attempts to extract or review data. These protocols make sure that no unauthorized devices, users, or services can access the network data. Primarily, these protocols depend on encryption to secure data.

1. Secure Socket Layer(SSL)
2. Hypertext Transfer Protocol(HTTPS)

# DNS (Domain name system)

DNS is a hierarchical and distributed naming system that translates domain names into IP addresses. When you type a domain name like www.example.org into your browser, DNS ensures that the request reaches the correct server by resolving the domain to its corresponding IP address.

# Working DNS


Step 01: User requests and local cache is checked
Computer browser requests to visit https://geeksforgeeks.org
Local Cache is checked for requested address

# Working DNS

**Step 02: Host Files are checked**

If IP address not found in cache

Checking Hosts File →

Hosts File

The hosts file is a system file that maps domain names to IP addresses locally

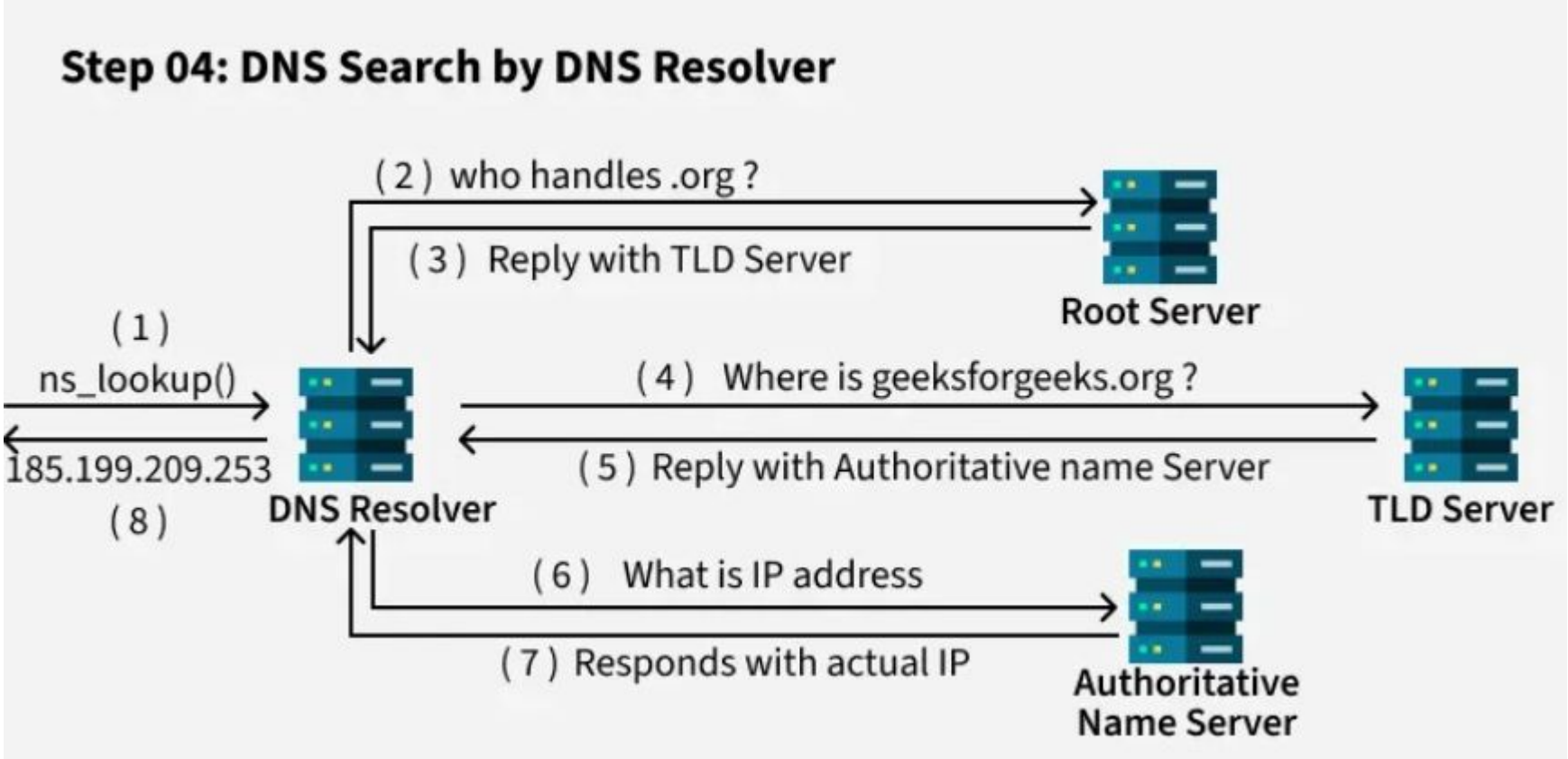
# Working DNS

**Step 03: Query DNS Resolver**

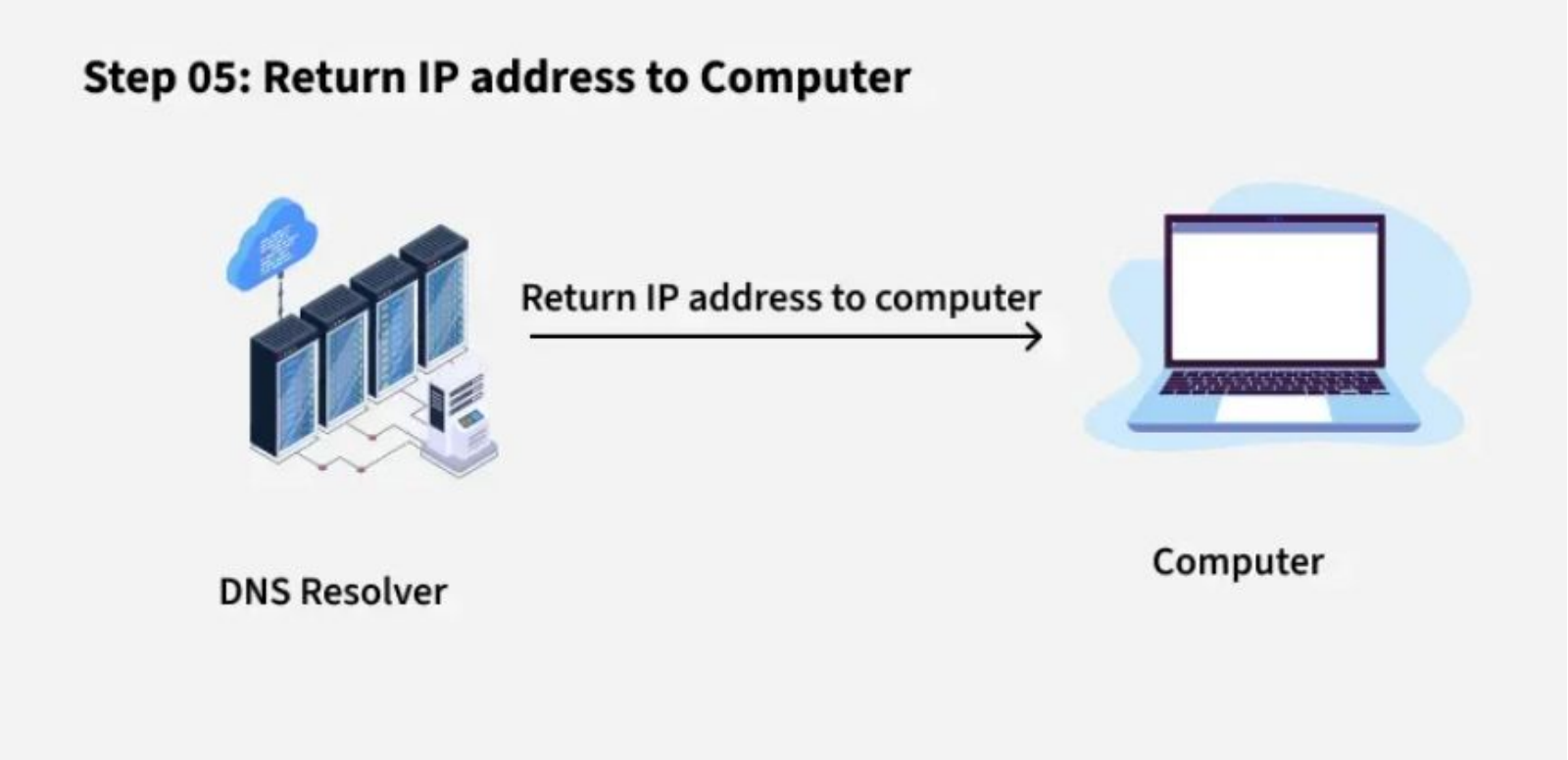
If IP address not found locally

DNS Resolver is a server provided by ISP

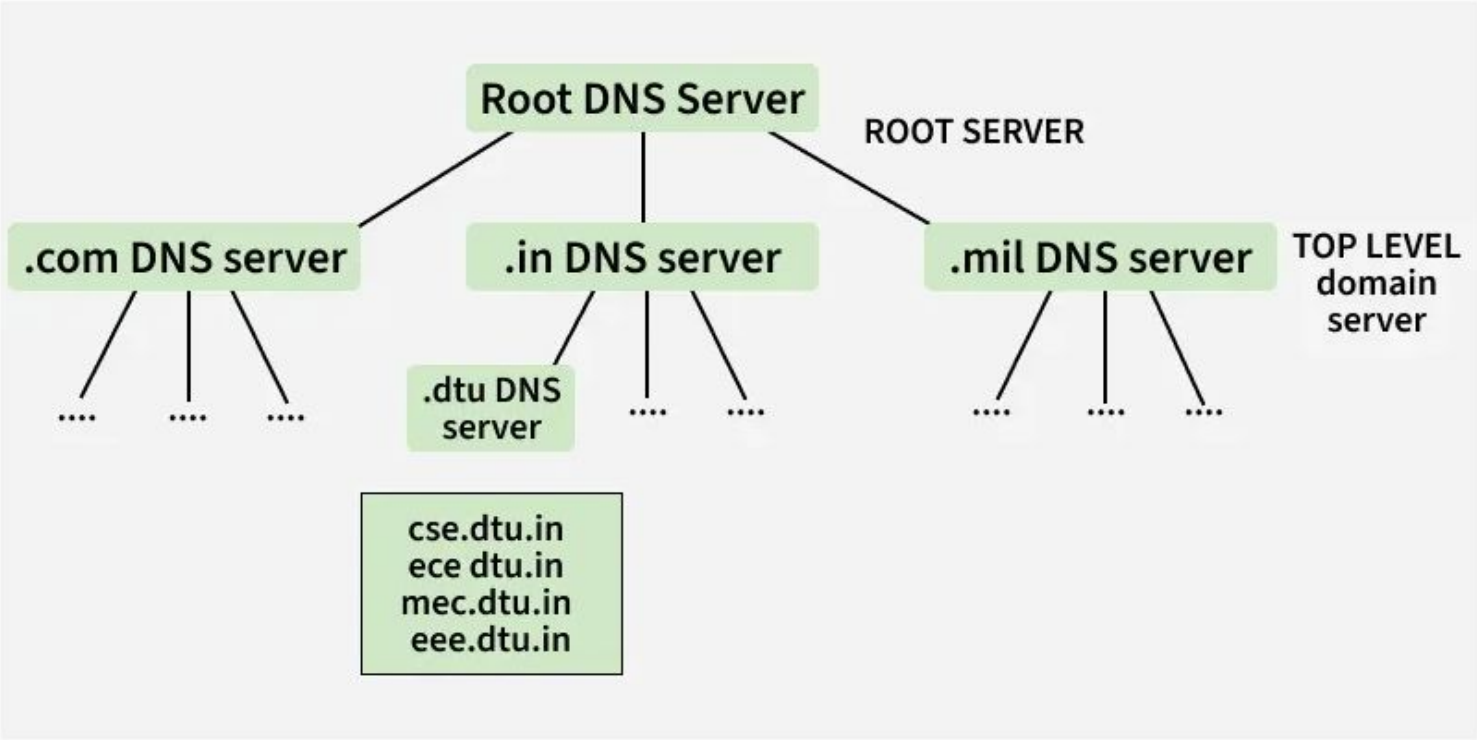
# Working DNS


Step 04: DNS Search by DNS Resolver
( 2 ) who handles .org ?
( 3 ) Reply with TLD Server
Root Server
( 1 )
ns_lookup()
185.199.209.253
( 8 )
DNS Resolver
( 4 ) Where is geeksforgeeks.org ?
( 5 ) Reply with Authoritative name Server
TLD Server
( 6 ) What is IP address
( 7 ) Responds with actual IP
Authoritative
Name Server

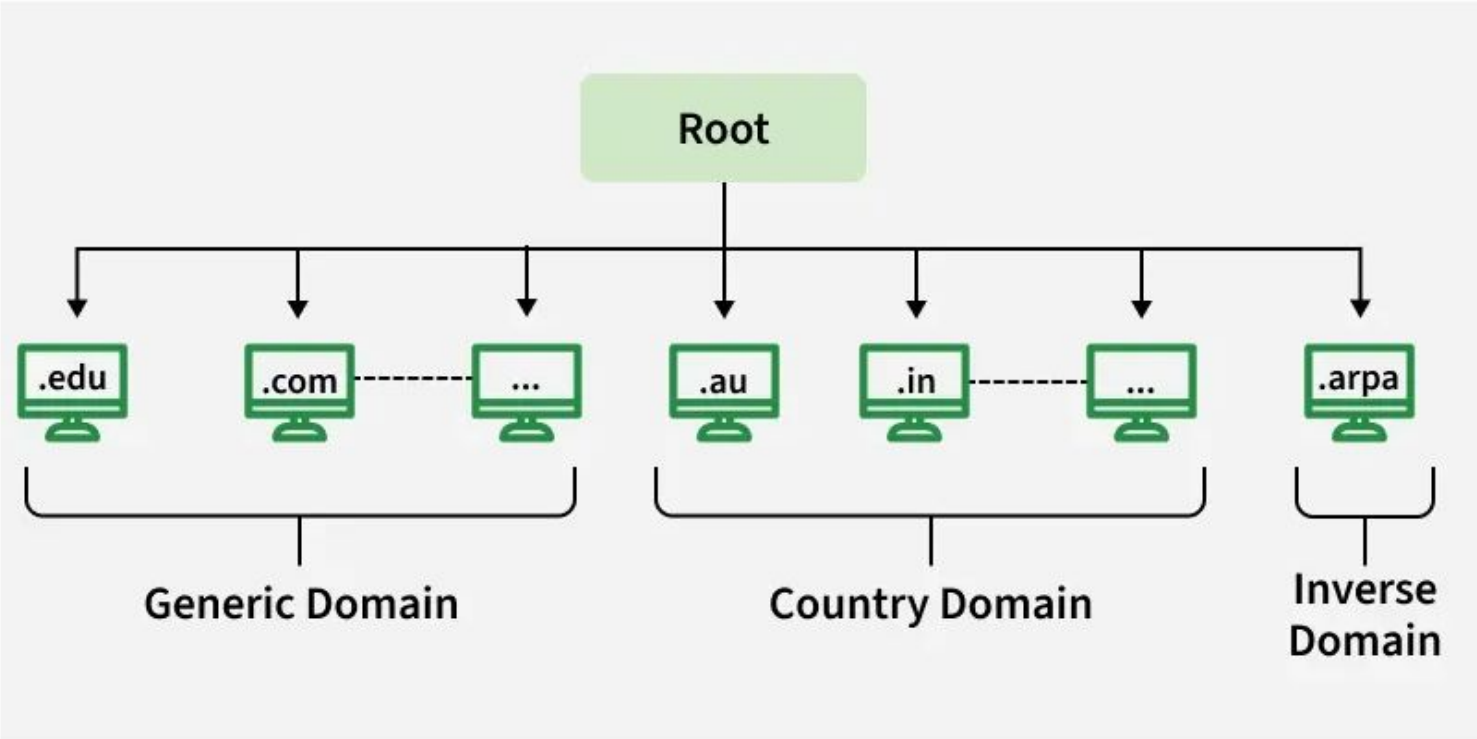
# Working DNS


Step 05: Return IP address to Computer
Return IP address to computer
DNS Resolver
Computer

# Structure of DNS



# Types of DNS


Root
.edu
.com
...
.au
.in
...
.arpa
Generic Domain
Country Domain
Inverse Domain